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Final Report

COOPERATIVE LOCALIZATION OF A WHEELED VEHICLE WITH GPS FAILURE
BASED ON RANGE MEASUREMENTS AND EKF

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1 Introduction

In the context of autonomous vehicles, being able to localize each vehicle precisely is predominant, even when an external signal is not available at any time. For instance, the GPS signal may not be available due to either a denied signal environment or a receiver failure. In the case of several wheeled vehicles moving together, they can work cooperatively to localize a vehicle with a GPS failure. A common technique is the triangulation based on ranging between the vehicle of interest and the others. Another is to directly take account the range measurements in an EKF. In this report, three methods are studied, based on RF (Radio Frequency) TOF (Time of Flight) ranging, magnetometer, and wheel speed rotary encoder.

2 Description of the studied system

A group of four wheeled vehicles that are moving together in two dimensions on the surface of earth is considered. The vehicles are differential wheeled robots, as shown in Fig. 1. The vehicles embed dead-reckoning sensors as well as position fixing sensors. Each vehicle has two incremental rotary encoders, one on each wheel, that provide directly angular velocity measurements. In order to correct the position obtained by the integration of the dead reckoning sensor measurements, the vehicles have access to a GPS signal at a low frequency, and a magnetometer to correct the attitude of the vehicle. Moreover, all of them have a RF Ranging system in order to determine the distance between each others. Finally, we suppose that they can communicate with each others and provide information such as their own position and attitude for example.

The situation that is studied in this report is represented on Fig. 1, in which one of the vehicle has lost his GPS signal, which will be localized cooperatively. The vehicles that have access to GPS positioning are designated by WV1 (Wheel Vehicle 1), WV2 and WV3. The vehicle of interest that doesn't have access to the GPS is designated by WV4. The frame of reference is a NED (North East Down) frame noted $\{t\} = \{O, \mathbf{n}, \mathbf{e}, \mathbf{d}\}$. As the vehicles evolve in two dimensions, the frame is given in 2D for convenience : $\{t\} = \{O, \mathbf{n}, \mathbf{e}\}$. A 2D body frame $\{b_i\} = \{O_i, \mathbf{x}_i, \mathbf{y}_i\}$, $i \in \{1, 2, 3, 4\}$, is fixed on each vehicle.

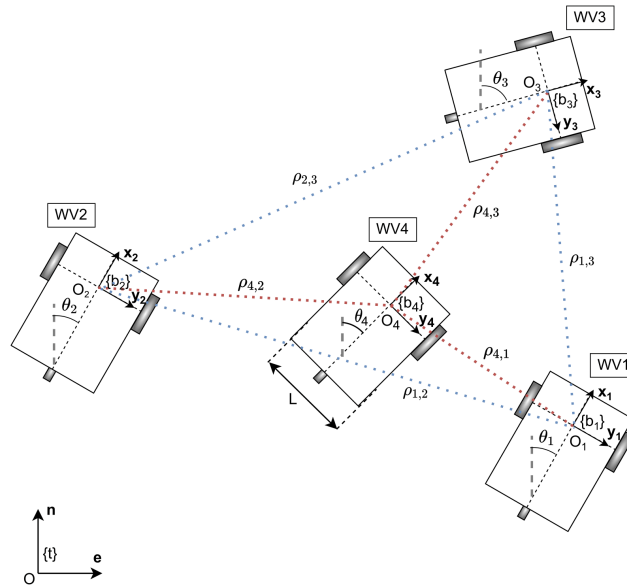


FIGURE 1 – Schematic view of the considered system.

3 Description of the considered approaches

Three different approaches are used to localize the vehicle with GPS failure (WV4). First, two decentralized approaches are considered, where each vehicle localizes itself by running its own algorithm. Then a centralized approach is considered. In the decentralized approaches, range measurements are made only between WV4 and others, whereas in the centralized approach the measurements are made between each vehicle.

3.1 Approach A : decentralized with triangulation pseudo position measurements

In this approach, the vehicle WV4 uses range measurements with the three other vehicles in a least square algorithm. The result is taken as a pseudo position fixing measurement in an EKF. The accuracy of these pseudo position measurements depends on the position of the three other vehicles, and therefore varies with time. The HDOP (Horizontal Dilution Of Precision) from the least square method is used to estimate the variance of the pseudo position measurement.

3.2 Approach B : decentralized with direct range measurements

In this second decentralized approach, the vehicle WV4 doesn't use pseudo position measurements but directly the range measurement in an EKF.

3.3 Approach C : centralized

The third approach uses centralized data from all the vehicles with one algorithm that is responsible to determine the position and orientation of each vehicle. In this case, the range measurements are not only performed between WV4 and the others, but between each vehicle.

4 Specification of expected performance

Better results with the centralized approach are expected, as the data can be manipulated more efficiently, and the range measurements are made between every vehicle. Between both decentralized methods, better results are expected for the one with direct range measurements (approach B) than with triangulation pseudo position measurement (approach A). Indeed, the bias of the pseudo position measurement model is not well defined and cannot be easily estimated. In a general way, an accuracy on position of about 10 m and an orientation accuracy of a few degrees are expected.

5 Sensors

5.1 Dead reckoning Sensors

The dead reckoning sensors used are wheel speed encoders. The model of the sensor is the following :

$$\tilde{\dot{\varphi}} = \dot{\varphi} + \eta_{\dot{\varphi}} \quad (1)$$

where $\dot{\varphi}$ is an angular velocity. We suppose that the noise $\eta_{\dot{\varphi}}$ is a CT (Continuous Time) WGN (White Gaussian Noise) and includes the measurement noise and the quantization noise from the encoder. We consider that its PSD (Power Spectral Density) is given by $\sigma_{\eta_{\dot{\varphi}}}^2 = 1 \text{ (rad.s}^{-1}\text{)}^2/\text{Hz}$.

As slip is introduced in the kinematics equations, as explain further in section 6.1.1, the measurements of the wheel speed must take that into account. We simulate the sensor measurements with the following equation :

$$\tilde{\varphi} = \left(\sqrt{(\dot{n} + \xi_n)^2 + (\dot{e} + \xi_e)^2} \pm \frac{L}{2}(\dot{\theta} + \xi_\theta) \right) \cdot \frac{1}{R + \delta R} + \eta_{\tilde{\varphi}} \quad (2)$$

The sign $\pm$ depend on which wheel is considered.

5.2 Magnetometer

The magnetometer measurements are modelled as follow :

$$\tilde{\mathbf{m}} = \mathbf{m} + \eta_m \quad (3)$$

where η_m is a DT (Discrete Time) WGN with covariance matrix $\sigma_m^2 \mathbf{I}_2$, $\sigma_m = 1 \mu\text{T}$.

The frame of the magnetometer is considered to be the same as the body frame $\{\mathbf{b}\}$ of the vehicle and will therefore depends on the orientation θ of the vehicle. We can write :

$$\tilde{\mathbf{m}}^b = \mathbf{m}^b(\theta) + \eta_m \quad (4)$$

As the frame $\{\mathbf{t}\}$ and $\{\mathbf{b}\}$ are in the same plane, only the 2D components of the magnetic vector are useful to deduce information on θ . The 2D magnetic field vector in $\{\mathbf{b}\}$ can be expressed as a rotation matrix times the 2D magnetic field vector in $\{\mathbf{t}\}$:

$$\mathbf{m}^b(\theta) = R_t^b(\theta) \mathbf{m}^t \quad (5)$$

with $\mathbf{m}^t = [m_n^t \ m_e^t]^T \approx [18098.6 \ -4472.6]^T$ nT at YUL Airport. Eq. 5 can then be written as :

$$\tilde{\mathbf{m}}^b(\theta) = R_t^b(\theta) \mathbf{m}^t + \eta_m^b \quad (6)$$

With R_t^b the rotation matrix to pass from the coordinates of $\{\mathbf{t}\}$ to the coordinates of $\{\mathbf{b}\}$:

$$R_t^b = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \quad (7)$$

Finally, the measurements are given by :

$$\mathbf{y}_k = \begin{bmatrix} \tilde{m}_x^b \\ \tilde{m}_y^b \end{bmatrix} = \begin{bmatrix} \cos(\theta)m_n^t + \sin(\theta)m_e^t \\ -\sin(\theta)m_n^t + \cos(\theta)m_e^t \end{bmatrix} + \begin{bmatrix} \nu_{m,x}^b \\ \nu_{m,y}^b \end{bmatrix} \quad (8)$$

This equation is used to simulate the measurements thanks to the orientation θ from the true state of the vehicle.

5.3 Global Positioning System (GPS)

The GPS receiver is modeled as :

$$\mathbf{y}_k = \begin{bmatrix} \tilde{n}_k \\ \tilde{e}_k \end{bmatrix} = \begin{bmatrix} n_k + b_{G,n,k} + \eta_{G,n,k} \\ e_k + b_{G,e,k} + \eta_{G,e,k} \end{bmatrix} \quad (9)$$

where $b_{G,n,k}$, $b_{G,e,k}$ are the bias for the coordinate n and e respectively. We supposed that they follow the dynamics given in Eq. (10) :

$$\begin{aligned} \dot{b}_{G,n} &= -\lambda_{G,n} \cdot b_{G,n} + w_{b_{G,n}} \\ \dot{b}_{G,e} &= -\lambda_{G,e} \cdot b_{G,e} + w_{b_{G,e}} \end{aligned} \quad (10)$$

where $\lambda_{G,n} = \lambda_{G,e} = 1/60$ are time constants. $w_{b_{G,n}}$ and $w_{b_{G,e}}$ are CT WGN with PSD $\sigma_{w_{b_{G,n}}}^2 = \sigma_{w_{b_{G,e}}}^2 = (\frac{3}{\sqrt{30}})^2 \text{ (m/s)}^2/\text{Hz}$.

$\eta_{G,n,k}$ and $\eta_{G,e,k}$ are DT WGN with variance $\sigma_{\eta_{G,n,k}}^2 = \sigma_{\eta_{G,e,k}}^2 = 1 \text{ m}^2$.

5.4 RF Ranging Sensor

The range measurement sensor follows the model defined as below :

$$\tilde{\rho}_{i,j} = \rho_{i,j} + c\delta t_{i,j} + \eta_{\rho}(\rho_{i,j}) \quad (11)$$

with $\rho_{i,j}$ is the distance between vehicles i and j. $\delta t_{i,j}$ is the clock shift between vehicles i and j, it is modeled by a second order linear system such that :

$$\begin{aligned} \dot{\delta t}_{i,j} &= f_{i,j} \\ \dot{f}_{i,j} &= w_{f_{i,j}} \end{aligned} \quad (12)$$

With $w_{f_{i,j}}$ a CT WGN with PSD $\sigma_{w_f}^2 = 0.1 \cdot 10^{-9} (\text{s}^{-1})^2/\text{Hz}$.

The additive noise $\eta_{\rho}(\rho_{i,j})$ is a DT WGN with variance $\sigma_{\eta_{\rho}}^2(\rho_{i,j})$ that depends on the distance between the emitter and the receiver as $\sigma_{\eta_{\rho}}(\rho_{i,j}) = 0.63 \cdot (\rho_{i,j})^{0.5} \text{ m}$. This equation corresponds to the linear interpolation of the data in the Figure 2 of the RF ranging sensor datasheet [1].

6 General system analysis

First, the kinematics and navigation equations of a wheeled vehicle are recalled. The hybrid extended Kalman filter used for sensor fusion is detailed.

6.1 Continuous-Time system's dynamics

6.1.1 Wheeled vehicle dynamics

In this part, the kinematics of a wheeled vehicle and the system's dynamics are studied. The kinematics state of the vehicle is given by its 2D coordinates in the frame $\{t\}$ and its orientation θ with respect to north axis $\mathbf{n}$.

The relevant parameters of the vehicle's geometry are the length $L = 50\text{cm}$ between the two wheels, and the radius of the left and right wheel, R_l and R_r respectively.

The kinematics equations of the vehicle are as follow :

$$\begin{cases} \dot{n} = \frac{1}{2} \cos(\theta)(R_l \dot{\varphi}_l + R_r \dot{\varphi}_r) \\ \dot{e} = \frac{1}{2} \sin(\theta)(R_l \dot{\varphi}_l + R_r \dot{\varphi}_r) \\ \dot{\theta} = \frac{1}{L}(R_l \dot{\varphi}_l - R_r \dot{\varphi}_r) \end{cases} \quad (13)$$

With $\dot{\varphi}_l$ and $\dot{\varphi}_r$ the angular speed of left and right wheels respectively.

These equations are true for an ideal system only, that is a slip-free rolling and a perfect knowledge of wheels speed and wheels radius. In order to take account of the error on the wheel radius, we express the radius as : $R_l = R_0 + \delta R_l$ and $R_r = R_0 + \delta R_r$, and add δR_l and δR_r in the state, with $R_0 = 10\text{cm}$. They represents the wheels radius variations principally due to the floor imperfections. Now, the kinematic equations can be written in term of measurable quantities $\tilde{\varphi}_l$ and $\tilde{\varphi}_r$ by inverting the sensor model (see 5). To make the system more realistic and take account for the slip, we add independent CT WGN noises ξ_n , ξ_e , ξ_θ , $\xi_{\delta R_l}$, $\xi_{\delta R_r}$. Their PSD are respectively $\sigma_{\xi_n}^2$, $\sigma_{\xi_e}^2$, $\sigma_{\xi_\theta}^2$, $\sigma_{\xi_{\delta R_l}}^2$ and $\sigma_{\xi_{\delta R_r}}^2$.

The system's CT dynamics can now be written as follow :

$$\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{w}, t) \implies \begin{cases} \dot{n} = \frac{1}{2} \cos(\theta)((R_0 + \delta R_l)(\tilde{\varphi}_l - \eta_{\dot{\varphi}_l}) + (R_0 + \delta R_r)(\tilde{\varphi}_r - \eta_{\dot{\varphi}_r})) + \xi_n \\ \dot{e} = \frac{1}{2} \sin(\theta)((R_0 + \delta R_l)(\tilde{\varphi}_l - \eta_{\dot{\varphi}_l}) + (R_0 + \delta R_r)(\tilde{\varphi}_r - \eta_{\dot{\varphi}_r})) + \xi_e \\ \dot{\theta} = \frac{1}{L}((R_0 + \delta R_l)(\tilde{\varphi}_l - \eta_{\dot{\varphi}_l}) - (R_0 + \delta R_r)(\tilde{\varphi}_r - \eta_{\dot{\varphi}_r})) + \xi_\theta \\ \delta \dot{R}_l = -\lambda_{\delta R_l} \delta R_l + \xi_{\delta R_l} \\ \delta \dot{R}_r = -\lambda_{\delta R_r} \delta R_r + \xi_{\delta R_r} \end{cases} \quad (14)$$

with the following minimal state :

$$\mathbf{x} = [n \quad e \quad \theta \quad \delta R_l \quad \delta R_r]^T \in \mathbb{R}^{5 \times 1}. \quad (15)$$

To write the navigation equations, the noises are set to zero and each element from the state is replaced

by its estimate. Then the navigation equations are given by :

$$\begin{cases} \dot{\hat{n}} = \frac{1}{2}\cos(\hat{\theta})((R_0 + \delta\hat{R}_l)\tilde{\varphi}_l + (R_0 + \delta\hat{R}_r)\tilde{\varphi}_r) \\ \dot{\hat{e}} = \frac{1}{2}\sin(\hat{\theta})((R_0 + \delta\hat{R}_l)\tilde{\varphi}_l + (R_0 + \delta\hat{R}_r)\tilde{\varphi}_r) \\ \dot{\hat{\theta}} = \frac{1}{L}((R_0 + \delta\hat{R}_l)\tilde{\varphi}_l - (R_0 + \delta\hat{R}_r)\tilde{\varphi}_r) \\ \delta\dot{\hat{R}}_l = -\lambda_{\delta R_l}\delta\hat{R}_l \\ \delta\dot{\hat{R}}_r = -\lambda_{\delta R_r}\delta\hat{R}_r \end{cases} \quad (16)$$

The navigation equations are integrated numerically to propagate the state estimate with high frequency wheel speed measurements between two discrete-time position fixing measurements.

6.1.2 GPS biases dynamics

In all approaches, the vehicles WV1,2,3 receive position fixing measurement thanks to GPS. As seen in section 5.3, the GPS have a bias for each coordinates. Those bias are unknown and have to be estimated. Therefore, they are included in the state of the corresponding vehicles in the different approaches. The corresponding state variables are $[b_{G,n} \ b_{G,e}]^T$. As defined in equation (10), the dynamics of the biases are given below with the corresponding navigation equations :

$$\begin{cases} \dot{b}_{G,n} = -\lambda_{G,n} \cdot b_{G,n} + w_{b_{G,n}} \\ \dot{b}_{G,e} = -\lambda_{G,e} \cdot b_{G,e} + w_{b_{G,e}} \end{cases} \implies \begin{cases} \dot{\hat{b}}_{G,n} = -\lambda_{G,n} \cdot \hat{b}_{G,n} \\ \dot{\hat{b}}_{G,e} = -\lambda_{G,e} \cdot \hat{b}_{G,e} \end{cases} \quad (17)$$

6.1.3 Clock shifts dynamics

In the 2nd and 3rd approach, we need to estimate the clock shifts due to the ranging sensor described in section 5.4. But as the GPS biases, each clock shift and its derivative have to be estimated. The corresponding state variables are $[\delta t_{i,j} \ f_{i,j}]^T$. Their kinematics equations are given in eq.(12) and repeated below with navigation equations :

$$\begin{cases} \dot{\delta t}_{i,j} = f_{i,j} \\ \dot{f}_{i,j} = w_{f_{i,j}} \end{cases} \implies \begin{cases} \dot{\hat{\delta t}}_{i,j} = \hat{f}_{i,j} \\ \dot{\hat{f}}_{i,j} = 0 \end{cases} \quad (18)$$

6.2 Hybrid Extended Kalman Filter

For each approach, the problem has a CT (Continuous-Time) nonlinear dynamics model $f(\mathbf{x}, \mathbf{w}, t)$ (14) and a DT (Discrete-Time) measurements model $h(\mathbf{x}_k, \mathbf{v}_k)$. Therefore a hybrid extended Kalman filter is used for sensor fusion and is detailed in this section. Both approach A and B use four hybrid EKF (one for each vehicle) and the approach C uses one EKF for the overall state. The specific states and DT measurements for each approach and vehicle are detailed in section 7.

6.2.1 DT Kalman update

For each DT measurement, the observation matrix $\mathbf{C} = \frac{\partial h}{\partial \mathbf{x}}|_{(\hat{\mathbf{x}}_k^-, \mathbf{0})}$, the matrix $\mathbf{D} = \frac{\partial h}{\partial \mathbf{v}}|_{(\hat{\mathbf{x}}_k^-, \mathbf{0})}$ and the DT noise measurement covariance matrix $\mathbf{N}_y$ such that $\mathbb{E}[\mathbf{v}_k \mathbf{v}_k^T] = \mathbf{N}_y \delta_{k-l}$ are calculated. For each approach, the matrices are given in annex A for vehicles with GPS (WV1,2,3) and the vehicle without GPS (WV4). Note some differences depending the considered sensor and approach.

The three matrices calculated just above are used in the calculation of the Kalman gain (19). The Kalman gain is then used to update and correct the covariance matrix Σ_k^+ and the state $\hat{\mathbf{x}}_k^+$ in equations (20).

$$\mathbf{K}_k = \Sigma_k^- \mathbf{C}_k^T (\mathbf{C}_k \Sigma_k^- \mathbf{C}_k^T + \mathbf{D}_k \mathbf{N}_y \mathbf{D}_k^T)^{-1} \quad (19)$$

$$\begin{aligned} \Sigma_k^+ &= (\mathbf{I} - \mathbf{K}_k \mathbf{C}_k) \Sigma_k^- \\ \hat{\mathbf{x}}_k^+ &= \hat{\mathbf{x}}_k^- + \mathbf{K}_k (\mathbf{y}_k - h(\hat{\mathbf{x}}_k^-, \mathbf{0})) \end{aligned} \quad (20)$$

6.2.2 CT Kalman propagation

After the Kalman update thanks to the DT measurements is done, the state has to be propagated with the CT dynamics and the dead reckoning sensor measurements.

The kinematics equations are linearized around $\mathbf{x} = \bar{\mathbf{x}}$ and $\mathbf{w} = 0$ to be able to propagate the error covariance matrix :

$$\mathbf{x} = \bar{\mathbf{x}} + \mathbf{A}(t)(\mathbf{x} - \bar{\mathbf{x}}) + \mathbf{B}(t)\mathbf{w}(t) \quad (21)$$

with $\mathbf{A}(t) = \frac{\partial f}{\partial \mathbf{x}} \Big|_{(\bar{\mathbf{x}}(t), \mathbf{0}, t)}$ and $\mathbf{B}(t) = \frac{\partial f}{\partial \mathbf{w}} \Big|_{(\bar{\mathbf{x}}(t), \mathbf{0}, t)}$. But $\bar{\mathbf{x}}$ is a true trajectory unavailable, then we choose $\bar{\mathbf{x}} = \hat{\mathbf{x}}$. Explicit matrices $\mathbf{A}(t)$ and $\mathbf{B}(t)$ are given again in annex A.

To propagate the covariance matrix between a time t_{k-1} and t_k , the differential equation 22 is discretized, leading to the equation 23.

$$\dot{\Sigma} = \mathbf{A}(t)\Sigma + \Sigma\mathbf{A}(t)^T + \mathbf{B}(t)\mathbf{Q}(t)\mathbf{B}(t)^T \quad (22)$$

By consider a time increment $\tau_{i+1} - \tau_i$ with $t_{k-1} = \tau_0 < \tau_1 < \dots < \tau_i < \dots < \tau_n = t_k$

$$\begin{aligned} \Sigma_0 &= \Sigma_{k-1}^+ \\ \rightarrow \Sigma_{i+1} &= \Phi_i \Sigma_i \Phi_i^T + \mathbf{N}_i \quad (i = 0, \dots, n-1) \\ \rightarrow \Sigma_k^- &= \Sigma_n \end{aligned} \quad (23)$$

with

$$\Phi_i = e^{\mathbf{A}_i(\tau_{i+1} - \tau_i)} \quad (24)$$

and

$$\begin{aligned} \mathbf{N}_i &= \int_0^{\tau_{i+1} - \tau_i} e^{\mathbf{A}_i t} \mathbf{B}_i \mathbf{Q}_i \mathbf{B}_i^T e^{\mathbf{A}_i^T t} dt \\ &\approx (\tau_{i+1} - \tau_i) \mathbf{B}_i \mathbf{Q}_i \mathbf{B}_i^T \quad (\tau_{i+1} - \tau_i \ll 1) \end{aligned} \quad (25)$$

where $\mathbf{Q}$ is the covariance matrix of noises of the CT dynamics. Explicit form of $\mathbf{Q}$ is given in annex A.

To initialized the algorithm :

$$\hat{\mathbf{x}}_0^+ = \mathbb{E}[\mathbf{x}_0], \quad \Sigma_0^+ = \mathbb{E}[(\mathbf{x}_0 - \hat{\mathbf{x}}_0^+)(\mathbf{x}_0 - \hat{\mathbf{x}}_0^+)^T] \quad (26)$$

7 Approaches analysis

7.1 Approach A : decentralized with triangulation pseudo position measurements

7.1.1 Vehicles' states

The state of the vehicle WV4 is simply the state given by the dynamics of a wheeled vehicle described in Eq. (15) and is repeated here :

$$\mathbf{x}_4 = [n_4 \quad e_4 \quad \theta_4 \quad \delta R_{l,4} \quad \delta R_{r,4}]^T \in \mathbb{R}^{5 \times 1}. \quad (27)$$

The vehicles WV1, WV2 and WV3 have access to GPS measurements. Therefore, their states are augmented states that include the GPS bias, whose dynamics are described in Eq.(10) :

$$\mathbf{x}_i = [n_i \quad e_i \quad \theta_i \quad \delta R_{l,i} \quad \delta R_{r,i} \quad b_{Gni} \quad b_{Gei}]^T \in \mathbb{R}^{7 \times 1}. \quad (i = 1, \dots, 3) \quad (28)$$

Then for each vehicle, the dynamics and navigation equations are composed with equations from (14) to (18).

7.1.2 Triangulation algorithm

The goal of the triangulation is to give an estimate of the position of WV4 thanks to range measurements between WV4 and others, and their position estimates.

Basic Levenberg-Marquardt algorithm : A basic Levenberg-Marquardt algorithm is implemented to find $\tilde{n}_4$ and $\tilde{e}_4$ of WV4 that minimize the following least-square optimization problem :

$$\begin{bmatrix} \tilde{n}_4 \\ \tilde{e}_4 \end{bmatrix} = \underset{n,e}{\operatorname{argmin}} \sum_{i=1}^3 \left| \operatorname{norm} \left(\begin{bmatrix} n \\ e \end{bmatrix} - \begin{bmatrix} \hat{n}_i \\ \hat{e}_i \end{bmatrix} \right) - \tilde{\rho}_{4i} \right|^2 \quad (29)$$

The range measurements are $\tilde{\rho}_4 = [\tilde{\rho}_{4,1}, \tilde{\rho}_{4,2}, \tilde{\rho}_{4,3}]^T$ and the estimates of the positions of WV1-2-3 are $\hat{\mathbf{b}} = \begin{bmatrix} \hat{n}_1 & \hat{n}_2 & \hat{n}_3 \\ \hat{e}_1 & \hat{e}_2 & \hat{e}_3 \end{bmatrix}$ respectively. The algorithm is initialized with $\begin{bmatrix} n \\ e \end{bmatrix}^T = [0 \quad 0]^T$.

Then the ranges $\hat{\rho}_4 = [\hat{\rho}_{4,1}, \hat{\rho}_{4,2}, \hat{\rho}_{4,3}]^T$ between the actual position estimate and the position estimate of the other vehicles are determined with :

$$\hat{\rho}_{4,i} = \operatorname{norm} \left(\begin{bmatrix} n \\ e \end{bmatrix} - \begin{bmatrix} \hat{n}_i \\ \hat{e}_i \end{bmatrix} \right). \quad (30)$$

The new estimate of position measurement is given by :

$$\begin{bmatrix} n \\ e \end{bmatrix}^{k+1} = \begin{bmatrix} n \\ e \end{bmatrix}^k - (\mathbf{J}_G^T \mathbf{J}_G + \operatorname{diag}(\mathbf{J}_G^T \mathbf{J}_G))^{-1} \mathbf{J}_G^T (\hat{\rho}_4 - \tilde{\rho}_4) \quad (31)$$

with $\mathbf{J}_G$ the Jacobienne of $\hat{\rho}_4$:

$$\mathbf{J}_G = \begin{bmatrix} \frac{n-\hat{n}_1}{\hat{\rho}_{4,1}} & \frac{e-\hat{e}_1}{\hat{\rho}_{4,1}} \\ \frac{n-\hat{n}_2}{\hat{\rho}_{4,2}} & \frac{e-\hat{e}_2}{\hat{\rho}_{4,2}} \\ \frac{n-\hat{n}_3}{\hat{\rho}_{4,3}} & \frac{e-\hat{e}_3}{\hat{\rho}_{4,3}} \end{bmatrix} \quad (32)$$

According to our tests, the algorithm converge after 50 loops. Therefore, the iterative equations (30) to (32) are computed 50 times and the final value is taken as pseudo position measurement for WV4.

Output : The algorithm output is considered as virtual position measurement and could be defined modelled as follow in a general way :

$$\begin{bmatrix} \tilde{n}_k \\ \tilde{e}_k \end{bmatrix} = \begin{bmatrix} n_k \\ e_k \end{bmatrix} + \begin{bmatrix} b_{\Delta,n,k} \\ b_{\Delta,e,k} \end{bmatrix} + \begin{bmatrix} \eta_{\Delta,n,k} \\ \eta_{\Delta,e,k} \end{bmatrix} \quad (33)$$

where $b_{\Delta,n,k}$, $b_{\Delta,e,k}$ are the bias for the coordinate n and e respectively. $[\eta_{\Delta,n,k} \ \eta_{\Delta,e,k}]^T$ is DT WGN with covariance matrix Σ_{Δ} .

The bias resulting from an numerical algorithm is not easy to estimate with respect to the input data of the algorithm. As an approximation, the biases $b_{\Delta,n,k}$ and $b_{\Delta,e,k}$ are set to zero. The covariance matrix of the algorithm output error can be estimated by the DOP :

$$\Sigma_{\Delta} = (J_g^T J_g)^{-1} J_g^T \Sigma_{\rho} J_g (J_g^T J_g)^{-1} \quad (34)$$

with Σ_{ρ} the covariance matrix of the range measurements :

$$\Sigma_{\rho} = \text{diag}([\sigma_{\rho 4,1}^2, \sigma_{\rho 4,2}^2, \sigma_{\rho 4,3}^2]) \quad (35)$$

With $\sigma_{\rho 4,i}$ the variance due to the measurement sensor noise but also to the incertitude on position from the vehicle i. It's formula is given in the following equation where $\Sigma_i(1 : 2, 1 : 2)$ approximate the incertitude of position of vehicle i because we don't know the orientation of the range measurement.

$$\sigma_{\rho 4,i}^2 = \sigma_{\eta \rho}^2(\rho_{i,j}) + \text{trace}(\Sigma_i(1 : 2, 1 : 2)) \quad (36)$$

7.1.3 Discrete-Time Position Fixing measurements

For each vehicle, we consider two independent measurements and separate them in two different Kalman updates : one for the position measurement and another for the magnetometer measurement. In case the vehicle have access to GPS (WVi, i = 1,2,3), the position measurement will be done with GPS. In the other case, for WV4 a triangulation pseudo position measurement is used. We repeat here for each case the DT measurement model $\mathbf{y}_k = h(\mathbf{x}_k, \mathbf{v}_k)$.

As given in equation (8), the magnetic field measurements are given as :

$$\mathbf{y}_k = \begin{bmatrix} \tilde{m}_x^b \\ \tilde{m}_y^b \end{bmatrix} = \begin{bmatrix} \cos(\theta)m_n^t + \sin(\theta)m_e^t \\ -\sin(\theta)m_n^t + \cos(\theta)m_e^t \end{bmatrix} + \begin{bmatrix} \nu_{m,x}^b \\ \nu_{m,y}^b \end{bmatrix} \quad (37)$$

As given in equation (9), the position measurements by GPS are given as :

$$\mathbf{y}_k = \begin{bmatrix} \tilde{n}_k \\ \tilde{e}_k \end{bmatrix} = \begin{bmatrix} n_k + b_{G,n,k} \\ e_k + b_{G,e,k} \end{bmatrix} + \begin{bmatrix} \eta_{G,n,k} \\ \eta_{G,e,k} \end{bmatrix} \quad (38)$$

The triangulation pseudo position measurements are given as :

$$\mathbf{y}_k = \begin{bmatrix} \tilde{n}_k \\ \tilde{e}_k \end{bmatrix} = \begin{bmatrix} n_k \\ e_k \end{bmatrix} + \begin{bmatrix} \eta_{\Delta,n,k} \\ \eta_{\Delta,e,k} \end{bmatrix} \quad (39)$$

7.2 Approach B : decentralized with direct range measurements

In the second decentralized approach, WV4 is localized by using the range measurement with the others vehicles directly in an EKF, with no pseudo triangulation pseudo position measurements.

7.2.1 Vehicle's states

The state of the vehicle WV4 is simply the state given by the dynamics of a wheeled vehicle described in Eq. (15) with the clock shifts from the ranging measurements (Eq. 12) :

$$\mathbf{x}_4 = [n_4 \quad e_4 \quad \theta_4 \quad \delta R_{l,4} \quad \delta R_{r,4} \quad \delta t_{4,1} \quad f_{4,1} \quad \delta t_{4,2} \quad f_{4,2} \quad \delta t_{4,3} \quad f_{4,3}]^T \in \mathbb{R}^{11 \times 1}. \quad (40)$$

The states of WV1, WV2, WV3 are the same as in the decentralized approach with triangulation pseudo position measurements (eq. 28).

Finally for each vehicle, the kinematics and navigation equations depends from the state variables and are composed with equations from (14) to (18).

7.2.2 Discrete-Time Position Fixing measurements

The DT measurements for the EKF are the same as described in section 7.1.3 for the approach A, except that for WV4 the triangulation measurement model is replaced by the range measurement model directly. We supposed that the three range measurements are performed at the same time, therefore the DT range measurements model for WV4 is the following and is based on the equation (11) :

$$\mathbf{y}_k = \begin{bmatrix} \tilde{\rho}_{4,1} \\ \tilde{\rho}_{4,2} \\ \tilde{\rho}_{4,3} \end{bmatrix} = \begin{bmatrix} \rho_{4,1} \\ \rho_{4,2} \\ \rho_{4,3} \end{bmatrix} + \begin{bmatrix} c \cdot \delta t_{4,1} \\ c \cdot \delta t_{4,2} \\ c \cdot \delta t_{4,3} \end{bmatrix} + \begin{bmatrix} \eta_{\rho 4,1} \\ \eta_{\rho 4,1} \\ \eta_{\rho 4,1} \end{bmatrix} \quad (41)$$

Again as explain in the equation (36), $\eta_{\rho 4,i}$ is a DT WGN noise with variance $\sigma_{\rho 4,i}^2$ given by :

$$\sigma_{\rho 4,i}^2 = \sigma_{\eta_{\rho}}^2(\rho_{i,j}) + trace(\Sigma_i(1 : 2, 1 : 2)) \quad (42)$$

7.3 Approach C : centralized

As said in section 3.3, the range measurements are not only performed between WV4 and the others, but between each vehicle

7.3.1 Overall state

First of all, the overall system's state have to be defined. For each vehicle, a sub-state $\mathbf{X}_i$ could be written. The sub-states of WV1-2-3 are the same as in approach A and B :

$$\mathbf{X}_i = [n_i \quad e_i \quad \theta_i \quad \delta R_{l,i} \quad \delta R_{r,i} \quad b_{G_n,i} \quad b_{G_e,i}]^T \in \mathbb{R}^{7 \times 1}, \quad (i = 1, 2, 3). \quad (43)$$

The vehicle WV4 sub-state is simply the minimal state, repeated here :

$$\mathbf{X}_4 = [n_4 \quad e_4 \quad \theta_4 \quad \delta R_{l,4} \quad \delta R_{r,4}]^T \in \mathbb{R}^{5 \times 1}. \quad (44)$$

The lasts elements that have to be added in the state are the clock shifts, between each vehicle. Each clock shift and its derivative are added in the following sub-states :

$$\mathbf{X}_{\rho_{i,j}} = [\delta t_{i,j} \quad f_{i,j}]^T \in \mathbb{R}^{2 \times 1}, \quad (i, j = 1, \dots, 4 \mid i < j). \quad (45)$$

Then the overall state is the combination of the precedents sub-states (eq. 46) :

$$\mathbf{X} = [\mathbf{X}_1 ; \mathbf{X}_2 ; \mathbf{X}_3 ; \mathbf{X}_4 ; \mathbf{X}_{\rho_{1,2}} ; \mathbf{X}_{\rho_{1,3}} ; \mathbf{X}_{\rho_{1,4}} ; \mathbf{X}_{\rho_{2,3}} ; \mathbf{X}_{\rho_{2,4}} ; \mathbf{X}_{\rho_{3,4}}] \in \mathbb{R}^{38 \times 1}. \quad (46)$$

Finally for each vehicle, the kinematics and navigation equations depends from the state variables are composed with equations from (14) to (18).

8 Implementation on Matlab

First, a class "WV" which represents a wheeled vehicle has been implemented. The class contains methods that implement an EKF, such as propagation of the true state and the state estimate of the vehicle, and Kalman update. In the two decentralized approaches, the EKF for each vehicle is directly performed by the class itself. In the centralized approach, the class WV is only used to represent a vehicle and its sub-state, but the EKF is implemented independently as it is centralized.

8.1 Decentralized approaches

For both decentralized approach, the algorithm globally the same, the only difference is the DT measurement for WV4 (triangulation vs ranging). The vehicles are represented by an instance of class WV. For each increment of time we do the corresponding measurements, state propagation and update inside the class. The pseudo code of the algorithm is given in figure 2. The two main methods of the class WV, *CT_measurement* and *DT_measurement*, are presented just below in fig. 3 and 4 respectively.

```

- Vehicles' initialization (WV(1),WV(2),WV(3),WV(4))
-  $h_{odo} \leftarrow 1/f_{odo}$  %period between lecture for each sensor
-  $h_{mag} \leftarrow 1/f_{mag}$ ;  $h_{GPS} \leftarrow 1/f_{GPS}$ ;  $h_{rho} \leftarrow 1/f_{rho}$ 
- For n from (0) to ( $f_{GPS} \cdot T_{simu} - 1$ )
  - For n from (0) to ( $f_{mag}/f_{GPS} - 1$ )
    - For o from (0) to ( $f_{odo}/f_{mag} - 1$ )
      - time <- time +  $h_{odo}$ 
      - WV(i).CT_measurements( $h_{odo}$ , acc(i)) %For i = (1,...,4)
    - End For
  - WV(i).DT_measurement("Magneto") %For i = (1,...,4)
  - WV(i).DT_measurement("GPS") %For i = (1,2,3)
- End For
- WV(4).DT_measurement("Triangulation") %Or "Ranging" for the second approach
- End For

```

FIGURE 2 – Decentralized main function

The propagation of the states (true and estimate) is done in the method *CT_measurement*. The vehicle trajectory is controlled by the acceleration, so the true position is calculated by numerically integrating the true acceleration twice. The pseudo code is given in figure 3.

```

-  $\tilde{\Phi} \leftarrow \Phi + \sigma_{\tilde{\Phi}} / \sqrt{acqTime} * randn(2,1)$ 
- WV.EstimateStatePropagation( $\tilde{\Phi}, h_{odo}$ ) %by linearization of navigation equations
-  $[A, B, Q] \leftarrow ABQ\_Calculation(\tilde{\Phi})$ 
-  $\Phi \leftarrow expm(A \cdot h_{odo})$ 
-  $\Sigma \leftarrow \Phi \Sigma \Phi^T + h_{odo} B Q B^T$ 
- WV.ReccordData()
- WV.TrueStatePropagation( $h_{odo}$ ) %by linearization of kinematics equations

```

FIGURE 3 – Decentralized approach : *CT_measurements* method of the class WV

Then, the method $DT_measurement("Type")$ effectuates the Kalman update. The pseudo code of our implementation is shown below in figure 4. Note the difference between the two decentralized approaches, "Triangulation" or "Ranging".

- Case ("type measure")
 - "Angle" :
 - $\widetilde{\mathbf{m}}^b \leftarrow \mathbf{R}_n^b(\theta) \cdot \mathbf{m}^n + \sigma_m \cdot randn(2,1)$
 - $\widehat{\mathbf{m}}^b \leftarrow \mathbf{R}_n^b(\hat{\theta}) \cdot \mathbf{m}^n$
 - $[\mathbf{C}, \mathbf{D}, \mathbf{N}_y] \leftarrow CDN_y_Calculation()$
 - $\mathbf{K} \leftarrow \Sigma \mathbf{C}^T (\mathbf{C} \Sigma \mathbf{C}^T + \mathbf{D} \mathbf{N}_y \mathbf{D}^T)^{-1}$
 - $\widehat{\mathbf{X}} \leftarrow \widehat{\mathbf{X}} + \mathbf{K}(\widetilde{\mathbf{m}}^b - \widehat{\mathbf{m}}^b)$
 - "GPS" : % Only vehicles 1,2,3
 - $\widehat{\mathbf{GPS}} \leftarrow \mathbf{Pos} + \mathbf{b}_G + \sigma_G \cdot randn(2,1)$
 - $\widehat{\mathbf{GPS}} \leftarrow \widehat{\mathbf{Pos}} + \widehat{\mathbf{b}}_G$
 - $[\mathbf{C}, \mathbf{D}, \mathbf{N}_y] \leftarrow CDN_y_Calculation()$
 - $\mathbf{K} \leftarrow \Sigma \mathbf{C}^T (\mathbf{C} \Sigma \mathbf{C}^T + \mathbf{D} \mathbf{N}_y \mathbf{D}^T)^{-1}$
 - $\widehat{\mathbf{X}} \leftarrow \widehat{\mathbf{X}} + \mathbf{K}(\widehat{\mathbf{GPS}} - \widehat{\mathbf{GPS}})$
 - "Triangulation" : % Only vehicle 4
 - $\rho_{4,i} \leftarrow norm(\mathbf{Pos} - WV(i).Pos)$ % $\rho_4 = [\rho_{4,1}, \rho_{4,2}, \rho_{4,3}]^T$
 - $\widetilde{\rho}_{4,l} \leftarrow \rho_{4,i} + c\delta t_{4,i} + \sigma_\rho(\rho_{4,i}) \cdot randn$ % For $i = (1,2,3)$
 - $[\widetilde{\mathbf{Pos}}, \mathbf{J}] \leftarrow triangulate(\widetilde{\rho}_{4,l}, WV(i).Pos)$ % For $i = (1,2,3)$
 - $\sigma_\rho(\rho_{4,i}) \leftarrow \sigma_\rho(\rho_{4,i}) + \sqrt{trace(WV(i). \Sigma_{Pos})}$
 - $\Sigma_\rho \leftarrow diag([\sigma_\rho^2(\rho_{4,1}), \sigma_\rho^2(\rho_{4,2}), \sigma_\rho^2(\rho_{4,3})])$
 - $[\mathbf{C}, \mathbf{D}, \mathbf{N}_y] \leftarrow CDN_y_Calculation(\Sigma_\rho)$
 - $\mathbf{K} \leftarrow \Sigma \mathbf{C}^T (\mathbf{C} \Sigma \mathbf{C}^T + \mathbf{D} \mathbf{N}_y \mathbf{D}^T)^{-1}$
 - $\widehat{\mathbf{X}} \leftarrow \widehat{\mathbf{X}} + \mathbf{K}(\widetilde{\mathbf{Pos}} - \widehat{\mathbf{Pos}})$
 - "Ranging" : % Only vehicle 4
 - $\rho_{4,i} \leftarrow norm(\mathbf{Pos} - WV(i).Pos)$ % $\rho_4 = [\rho_{4,1}, \rho_{4,2}, \rho_{4,3}]^T$
 - $\widehat{\rho}_{4,l} \leftarrow norm(\widehat{\mathbf{Pos}} - WV(i).Pos)$
 - $\widetilde{\rho}_{4,l} \leftarrow \rho_{4,i} + c\delta t_{4,i} + \sigma_\rho(\rho_{4,i}) \cdot randn$ % For $i = (1,2,3)$
 - $\widehat{\rho}_{4,l} \leftarrow \widehat{\rho}_{4,l} + c\delta t_{4,l}$
 - $\sigma_\rho(\rho_{4,i}) \leftarrow \sigma_\rho(\rho_{4,i}) + \sqrt{trace(WV(i). \Sigma(1:2, 1:2))}$
 - $[\mathbf{C}, \mathbf{D}, \mathbf{N}_y] \leftarrow CDN_y_Calculation(\sigma_\rho(\rho_{4,i}))$
 - $\mathbf{K} \leftarrow \Sigma \mathbf{C}^T (\mathbf{C} \Sigma \mathbf{C}^T + \mathbf{D} \mathbf{N}_y \mathbf{D}^T)^{-1}$
 - $\widehat{\mathbf{X}} \leftarrow \widehat{\mathbf{X}} + \mathbf{K}(\widetilde{\rho}_4 - \widehat{\rho}_4)$

FIGURE 4 – Decentralized approach : $DT_measurements$ method of the class WV

8.2 Centralized approach

In this approach, the class WV is only used to represent the vehicles and contain the true states and the state estimates. All the calculations are done separately and correspond to the following pseudo code.

```

- Vehicles' initialization (WV(1),WV(2),WV(3),WV(4))
-  $h_{Odo} \leftarrow 1/f_{Odo}$  %period between lecture for each sensor
-  $h_{Mag} \leftarrow 1/f_{Mag}$ ;  $h_{GPS} \leftarrow 1/f_{GPS}$ ;  $h_{Rho} \leftarrow 1/f_{Rho}$ 
- For m from (0) to ( $f_{GPS} \cdot T_{simu} - 1$ )
  - For n from (0) to ( $f_{Mag}/f_{GPS} - 1$ )
    - For o from (0) to ( $f_{Odo}/f_{Mag} - 1$ )
      - time <- time +  $h_{Odo}$ ,
        % Continuous time measurements and
        % Propagation of state (true & estimate)
      -  $\tilde{\Phi} \leftarrow \Phi + \sigma_{\tilde{\Phi}}/\sqrt{acqTime} * randn(8,1)$ 
      - WV.EstimateStatePropagation( $\tilde{\Phi}, h_{Odo}$ ) %by linearization of navigation
        equations
      -  $[A, B, Q] \leftarrow ABQ\_Calculation(\tilde{\Phi})$ 
      -  $\Phi \leftarrow expm(A \cdot h_{Odo})$ 
      -  $\Sigma \leftarrow \Phi \Sigma \Phi^T + h_{Odo} B Q B^T$ 
      - ReccordData()
      - TrueStatePropagation( $h_{Odo}, acc$ )
    - End For
    % Discrete time magnetic field measurements
    -  $\tilde{m}^b \leftarrow R_n^b(\theta) \cdot m^n + \sigma_m \cdot randn(8,1)$ 
    -  $\hat{m}^b \leftarrow R_n^b(\hat{\theta}) \cdot m^n$ 
    -  $[C, D, N_y] \leftarrow CDN_y\_Calculation()$ 
    -  $K \leftarrow \Sigma C^T (C \Sigma C^T + D N_y D^T)^{-1}$ 
    -  $\hat{X} \leftarrow \hat{X} + K(\tilde{m}^b - \hat{m}^b)$ 

    % Discrete time GPS measurements
    -  $\tilde{GPS} \leftarrow Pos + b_G + \sigma_G \cdot randn(8,1)$ 
    -  $\hat{GPS} \leftarrow \hat{Pos} + \hat{b}_G$ 
    -  $[C, D, N_y] \leftarrow CDN_y\_Calculation()$ 
    -  $K \leftarrow \Sigma C^T (C \Sigma C^T + D N_y D^T)^{-1}$ 
    -  $\hat{X} \leftarrow \hat{X} + K(\tilde{GPS} - \hat{GPS})$ 
  - End For
  % Discrete time WV4 position measurements
  -  $\rho_{i,j} \leftarrow norm(WV(j).Pos - WV(i).Pos)$  %  $\rho = [\rho_{1,2}, \rho_{1,3}, \rho_{1,4}, \rho_{2,3}, \rho_{2,4}, \rho_{3,4}]^T$ 
  -  $\hat{\rho}_{i,j} \leftarrow norm(WV(j).\hat{Pos} - WV(i).\hat{Pos})$  % For i,j = (1,...,4 | i<j)
  -  $\hat{\rho}_{i,j} \leftarrow \rho_{i,j} + c\delta t_{i,j} + \sigma_{\rho}(\rho_{i,j}) \cdot randn$  % For i,j = (1,...,4 | i<j)
  -  $\hat{\rho}_{i,j} \leftarrow \hat{\rho}_{i,j} + c\delta \hat{t}_{i,j}$ 
  -  $[C, D, N_y] \leftarrow CDN_y\_Calculation(\sigma_{\rho}(\rho_{i,j}))$ 
  -  $K \leftarrow \Sigma C^T (C \Sigma C^T + D N_y D^T)^{-1}$ 
  -  $\hat{X} \leftarrow \hat{X} + K(\tilde{\rho} - \rho)$ 
- End For

```

FIGURE 5 – Centralized approach : main algorithm

9 Simulation parameters

First of all, the frequency of the sensors are given in the following table :

TABLE 1 – Sensor frequencies

	Odometry	Magnetometer	Range	GPS
frequency [Hz]	0.05	0.5	0.5	1

9.1 Vehicles trajectories

The vehicles trajectories are circular. WV4's trajectory has twice the radius of other vehicles' trajectories. Every vehicles turn in the clockwise with the same angular speed and start at the north of the circle. The speed of WV4 is 2 m.s⁻¹ and therefore the speed of WV1-2-3 is 1 m.s⁻¹. The following table contains the position of the center of each circle and its radius :

TABLE 2 – Trajectories parameters

	WV1	WV2	WV3	WV4
Radius[m]	25	25	25	50
Center[m]	[200 0] ^T	[-100 100√3] ^T	[-100 -100√3] ^T	[0 0] ^T

The vehicles thus take 157s to complete a lap. Finally, the trajectory is initialized using the following equation :

$$\begin{aligned}
 Pos(0)_i &= Center(i) + Radius \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\
 Vit(0)_i &= Radius(i) \cdot \omega \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\
 P\ddot{os}(0)_i &= Radius(i) \cdot \omega^2 \cdot \begin{bmatrix} -1 \\ 0 \end{bmatrix}
 \end{aligned} \tag{47}$$

Moreover, the vehicles are commanded by acceleration. We can then numerically integrate the acceleration to get the true velocity and the true position.

$$\begin{aligned}
 P\ddot{os}(t)_i &= Radius(i) \cdot \omega^2 \cdot \begin{bmatrix} -\cos(\omega \cdot t) \\ -\sin(\omega \cdot t) \end{bmatrix} \\
 P\dot{os}(t)_i &= P\dot{os}(t)_i + P\ddot{os}(t)_i \cdot h_{Od0} \\
 Pos(t)_i &= Pos(t)_i + P\dot{os}(t)_i \cdot h_{Od0} + \frac{1}{2} P\ddot{os}(t)_i \cdot h_{Od0}^2
 \end{aligned} \tag{48}$$

Because of the non-holonomic vehicle, the θ angle depends of the vehicle velocity as in the following equation :

$$\theta(t) = \arctan\left(\frac{e(t)}{n(t)}\right) \tag{49}$$

Below a figure represents the true trajectories of the vehicles :

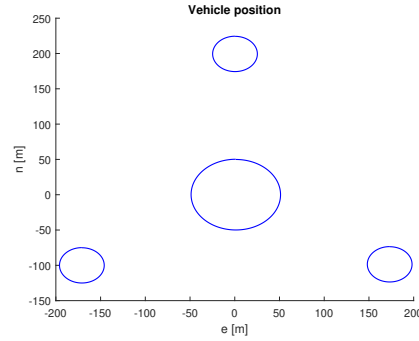


FIGURE 6 – True vehicles trajectories

9.2 Initialization of the states and covariances

For each state variable, an initialization has to be done. Then for the true state we choose some random values with mean and variance given in the following table :

TABLE 3 – Initialization parameters

Variables at $t = 0$	mean	variance
$\begin{bmatrix} n_i \\ e_i \end{bmatrix}$	$Pos(0)_i$ m	1^2 m ²
θ	$\pi/2$ rad	$(\pi/20)^2$ rad ²
δR	0 m	5^2 mm ²
b_G	0 m	$\sigma_{b,0}^2$ m ²
δt	0 s	$\sigma_{\delta t,0}^2$ s ²
f	0 s	$\sigma_{f,0}^2$

Then to initialize the estimate state, only means are taken. And the covariance matrix is initialized as a diagonal matrix with the corresponding covariance.

10 Results

10.1 WV4 VS WV1,2,3 comparison

For each approach, the distance error between the position estimate and the true position is computed for each vehicle. This is done for 100 MC (Monte Carlo) simulations, and the mean is computed. The figure (7), (9), and (10) show the mean of the distance error for WV4 and for WV1-2-3 (the mean of the distance error for the three vehicles is taken). In addition, the interval confidence at 95% are shown, that is 95% of statistical confidence that the distance error is below that value.

10.1.1 Approach A : decentralized with triangulation pseudo position measurements

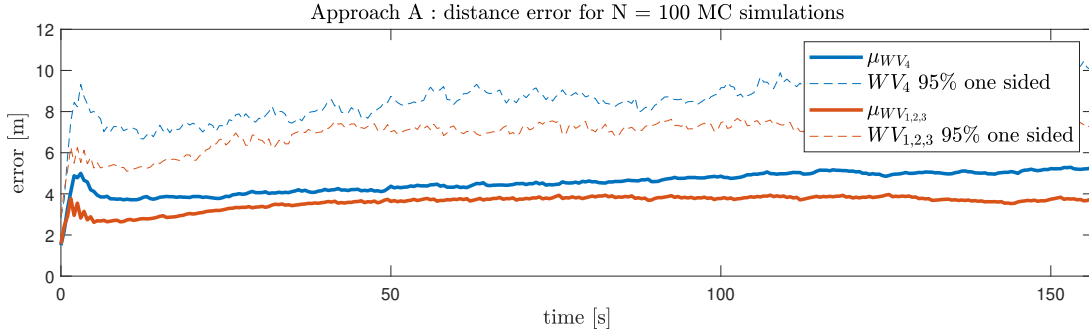


FIGURE 7 – Approach A : distance error for N=500 MC simulations, tuning = 40.

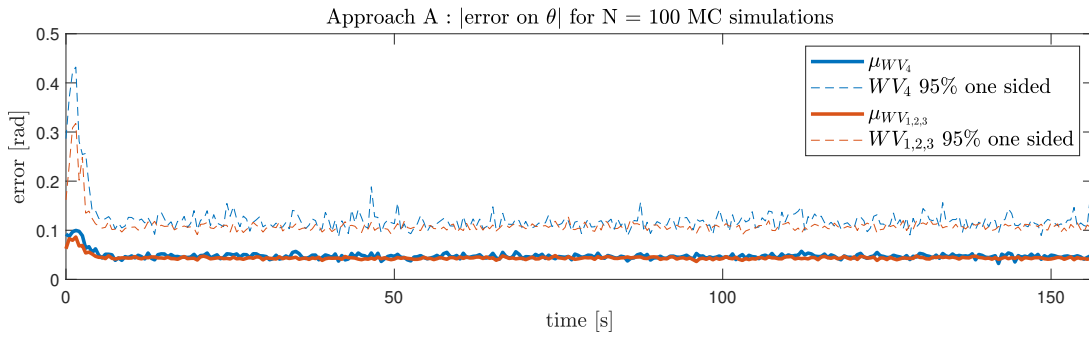


FIGURE 8 – Approach A : |error on θ | for N=500 MC simulations, tuning = 40.

10.1.2 Approach B : decentralized with direct range measurements

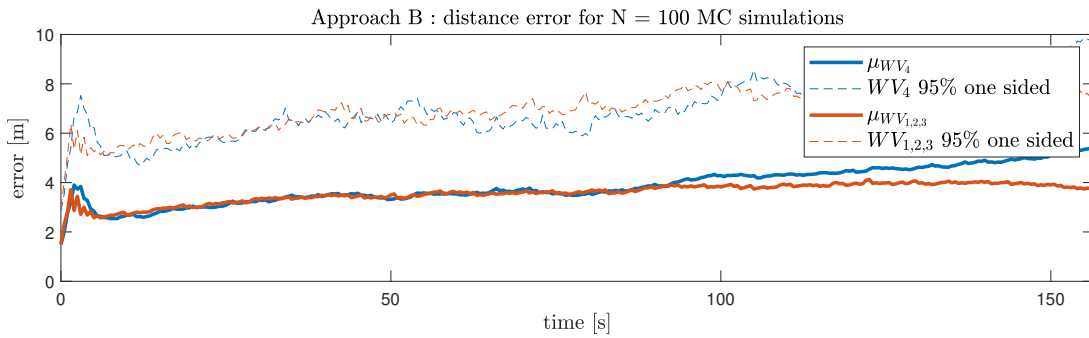


FIGURE 9 – Approach B : distance error for N=500 MC simulations, tuning = 40.

10.1.3 Approach C :

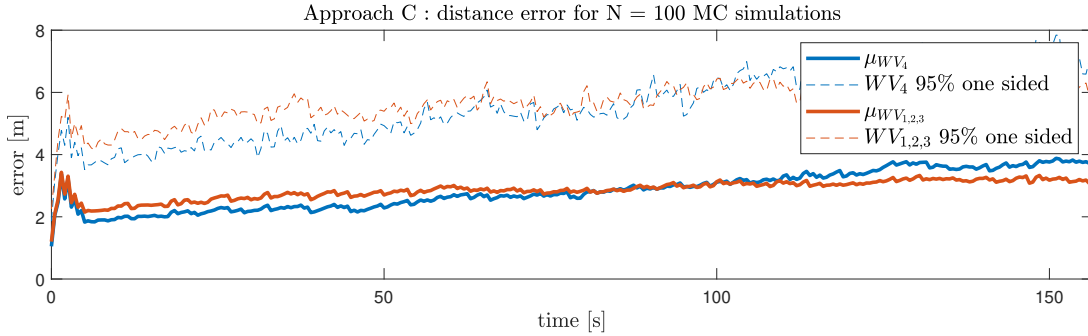


FIGURE 10 – Approach C : distance error for N=500 MC simulations, tuning = 40. £

We can see that for the approach A, the vehicle WV4 has a significantly higher distance error than the vehicles WV1-2-3. For approaches B and C, the difference between WV4 and WV1-2-3 is not significant. This could be explained by the fact that the bias resulting from the least square algorithm is not estimated, whereas in the two others methods, the clock shifts are estimated. For each approach, the distance error is growing with time, but very slowly. We can see that the distance error on WV4 is below 10m with 95% of statistical confidence, for all the approaches.

Then, we can see on Fig. 8 that the mean of the absolute error on the orientation is pretty small and is stable in time. Only the values for the approach A are shown, as the results on θ for approach B and C are very similar. This is explain by the fact that the orientation is corrected in the same way by the magnetometer for the three approaches. The comparison between the three approaches with respects to specific performance criteria is done in section 10.3.

10.2 NEES test for filter consistency

10.2.1 Details of the test

The NEES test tells if a considered filter respect the Gaussian hypothesis, that is

$$x_k | y_{0:k} \sim (\hat{x}_k, \Sigma_k) \quad (50)$$

We save the state error e_k and covariance matrix Σ_k after each Kalman update (after magnetometer measurements in our algorithm). Then it's possible after the Kalman update to calculate the NEES (Normalized (state) Estimation Error Squared) as follow :

$$\eta_k = e_k^T \Sigma_k^{-1} e_k \quad (51)$$

If the filter is consistent, we should have η_k that follows a χ_n^2 distribution, with n the state dimension. We need to decide if the filter is consistent after N simulation. In our case we chose $N = 100$. Now a chi-squared test is needed to identify if the NEES η_k respect the χ_n^2 distribution. We will accept the consistency if the average NEES $\bar{\eta}_k = \frac{1}{N} \sum_{i=1}^N \eta_k^i$ is in the acceptance following interval $[a_{1N}, a_{2N}]$. We choose the limits as $P(\bar{\eta}_k \in \mathbb{A}_N) = 1 - \alpha$ with $\alpha = 1\%$. Then :

$$\begin{cases} a_{1N} = \frac{1}{N} Q_{Nn} \left(\frac{\alpha}{2} \right) \\ a_{2N} = \frac{1}{N} Q_{Nn} \left(1 - \frac{\alpha}{2} \right) \end{cases} \quad (52)$$

With Q_{Nn} the quantile function for χ_{Nn}^2 .

10.2.2 Results

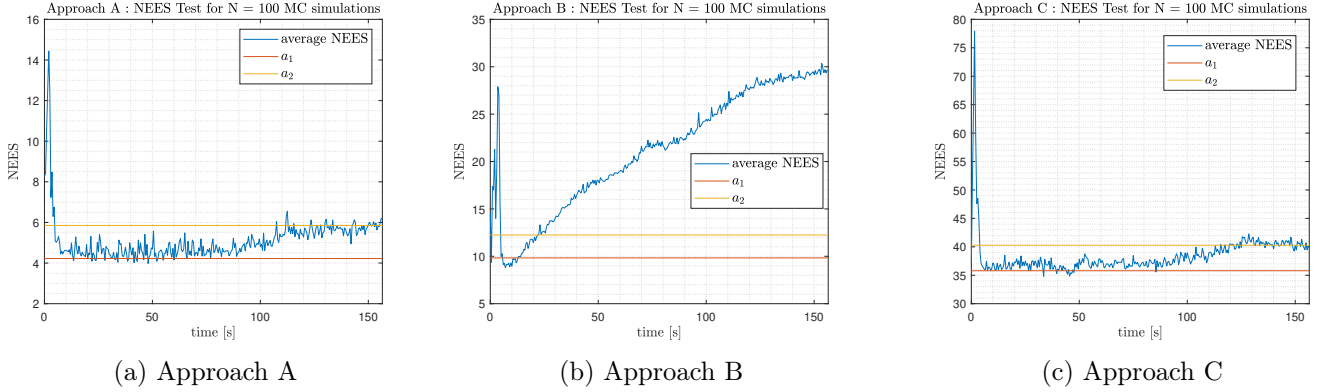


FIGURE 11 – NEES results with 100 Monte-Carlo simulations

In the above figures, we see that approach A and C globally fit in the acceptance interval limits. For both around 5% of values are out of the interval. However as described in the previous section, the limits were chosen for $\alpha = 1\%$. This why we cannot confirm that for these approaches the Kalman filter is consistent.

For approach 2, the graph seems really different and we honestly didn't understand why. Our first hypothesis is that there is an error in our code. We tried the same test without ranging measurement to localize WV4 and it was fitting in the limits. Therefore we think the error comes from the variance given for ranging measurements. We discussed as in equation (42) that the variance due to the ranging measurement was depending on the range sensor noise variance in addition to the position uncertainty of the vehicle with which the measurement is done. The equation is repeated below.

$$\sigma_{\rho 4,i}^2 = \sigma_{\eta \rho}^2 (\rho_{i,j}) + \text{trace}(\Sigma_i(1:2, 1:2)) \quad (53)$$

We used $\text{trace}(\Sigma_i(1:2, 1:2)) = \sigma_{n,i}^2 + \sigma_{e,i}^2$ to be sure that the added uncertainty was big enough, but maybe this is not the case.

Finally a last thing we observe with the NEES tests is that all graphs have a pic at the beginning. This is not recurrent, and it could mean that 100 Monte-Carlo simulations are not enough. As the system is quite complex, the simulations take a lot of time, about two hours to perform a NEES test.

10.3 Approaches comparison

The objective is to compare our three approaches with respects to a specific performance criterion. The two performance criteria are the following : distance error and orientation error.

10.3.1 Details of the test

We will test the approaches in pairs to see which offers the best performances. For each MC simulations i , and for both approaches we computed a performance criterion $C_i^{(1)}$ and $C_i^{(2)}$. It is the mean of the distance error or the mean of the absolute orientation error. Then we calculate the mean value over the N MC simulations of the difference of this criterion between algorithms as follow :

$$\bar{\Delta} = \frac{1}{N} \sum_{i=1}^N C_i^{(2)} - C_i^{(1)} \quad (54)$$

The empiric standard deviation with empiric mean is $\sigma_{\bar{\Delta}}^2 = \frac{1}{N} \sum_{i=1}^N (\Delta_i - \bar{\Delta})^2$. Then we declare that the approach (1) is better than approach (2) if :

$$\frac{\bar{\Delta}}{\sigma_{\bar{\Delta}}} > \mu_0 \quad (55)$$

with $\mu_0 = 1.64$.

10.3.2 Results

For each pair of approaches, we performed a comparison and the results are given in the table ??.

TABLE 4 – Results of the comparison tests

Filters :	A VS B		A VS C		B VS C	
Criteria :	d [m]	θ [rad]	d [m]	θ [rad]	d [m]	θ [rad]
$\bar{\Delta}$	-0.68	$-5.29 \cdot 10^{-5}$	-1.91	$-7.03 \cdot 10^{-5}$	-1.22	$-1.74 \cdot 10^{-5}$
$\sigma_{\bar{\Delta}}$	0.22	$3.84 \cdot 10^{-4}$	0.20	$4.19 \cdot 10^{-4}$	0.15	$3.72 \cdot 10^{-4}$
$\bar{\Delta}/\sigma_{\bar{\Delta}}$	-3.14	-0.13	-9.51	-0.17	-8.04	$-0.05 \cdot 10^{-5}$

For the distance error criterion, we can conclude that the approach C is better than approaches A and B. The approach B is better than the approach A. This confirm our initial hypothesis that the centralized approach would have better results, since the range measurements are made between each vehicle and the data can be manipulated more efficiently in terms of filtering performance. For the orientation criterion, we cannot say that one approach is significantly better than another. And that's because magnetometer measurements are done independently for each vehicles. It's a common point for all approaches.

11 Conclusion

All the proposed approaches give a reasonably good estimate of the true position of the vehicle WV4 with a GPS failure. The distance error is below 10 meters with 95% confidence for all the approaches, which was our initial expectation. Same, the orientation error is below 9 degrees with 95% confidence. The NEES test for filter consistency is not validated for the second decentralized approach, probably due to an error in the code. However, the decentralized approach with triangulation and the centralized approach are almost in the interval limits of the NEES tests. The centralized approach gives significantly better results in term of distance error compared to the to others. Between the two decentralized approach, the one with the triangulation gives significantly worst result compared to the other decentralized method. This is due to the fact that the biases resulting from the triangulation algorithm is not estimated.

Finally this project shows that there is some solutions to localize a vehicle with GPS failure with the same error as with a GPS. To do that the cooperative localization seems to be a good solution. Now we can ask us is that possible to combine the GPS measurement with cooperative navigation to improve the localization accuracy of the vehicles ?

Note : We allow the professor J.Le Ny to share this document to future classes.

A Kalman matrices

A.1 Decentralised approach with triangulation

A.1.1 Vehicles WV1,2,3 with GPS

Magnetic field measurements

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & -\sin(\hat{\theta})m_n^t + \cos(\hat{\theta})m_e^t & 0 & 0 & 0 & 0 \\ 0 & 0 & -\cos(\hat{\theta})m_n^t - \sin(\hat{\theta})m_e^t & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{D} = \mathbf{I}_2, \quad \mathbf{Ny} = \sigma_m^2 \cdot \mathbf{I}_2 \quad (56)$$

GPS measurements

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{D} = \mathbf{I}_2, \quad \mathbf{Ny} = \sigma_G^2 \cdot \mathbf{I}_2 \quad (57)$$

Continuous time propagation

$$\mathbf{A}(t) = \begin{bmatrix} 0 & 0 & -\hat{u} \sin(\hat{\theta}) & \frac{1}{2} \cos(\hat{\theta}) \tilde{\varphi}_l & \frac{1}{2} \cos(\hat{\theta}) \tilde{\varphi}_r & 0 & 0 \\ 0 & 0 & \hat{u} \cos(\hat{\theta}) & \frac{1}{2} \sin(\hat{\theta}) \tilde{\varphi}_l & \frac{1}{2} \sin(\hat{\theta}) \tilde{\varphi}_r & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{L} \tilde{\varphi}_l & -\frac{1}{L} \tilde{\varphi}_r & 0 & 0 \\ 0 & 0 & 0 & -\lambda_{\delta R_l} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\lambda_{\delta R_r} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\lambda_{b_{Gn}} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\lambda_{b_{Ge}} \end{bmatrix} \quad (58)$$

$$\hat{u} = \frac{1}{2}((R_l + \delta \hat{R}_l) \tilde{\varphi}_l + (R_r + \delta \hat{R}_r) \tilde{\varphi}_r)$$

$$\mathbf{B}(t) = \begin{bmatrix} -\frac{1}{2} \cos(\hat{\theta})(R_0 + \delta \hat{R}_l) & -\frac{1}{2} \cos(\hat{\theta})(R_0 + \delta \hat{R}_r) & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{2} \sin(\hat{\theta})(R_0 + \delta \hat{R}_l) & -\frac{1}{2} \sin(\hat{\theta})(R_0 + \delta \hat{R}_r) & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{L}(R_0 + \delta \hat{R}_l) & -\frac{1}{L}(R_0 + \delta \hat{R}_r) & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (59)$$

$$\hat{u} = \frac{1}{2}((R_l + \delta \hat{R}_l) \tilde{\varphi}_l + (R_r + \delta \hat{R}_r) \tilde{\varphi}_r)$$

$$\mathbf{Q} = \text{diag}([\sigma_{\varphi}^2, \sigma_{\varphi}^2, \sigma_n^2, \sigma_e^2, \sigma_{\theta}^2, \sigma_{\delta R}^2, \sigma_{\delta R}^2, \sigma_{b_G}^2, \sigma_{b_G}^2]), \quad \text{such as } \mathbb{E}[\mathbf{w}_k \mathbf{w}_l^T] = \mathbf{Q} \delta_{k-l} \quad (60)$$

A.1.2 Vehicle WV4 without GPS

Magnetic field measurements

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & -\sin(\hat{\theta})m_n^t + \cos(\hat{\theta})m_e^t & 0 & 0 \\ 0 & 0 & -\cos(\hat{\theta})m_n^t - \sin(\hat{\theta})m_e^t & 0 & 0 \end{bmatrix}, \quad \mathbf{D} = \mathbf{I}_2, \quad \mathbf{Ny} = \sigma_m^2 \cdot \mathbf{I}_2 \quad (61)$$

triangulation measurements

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{D} = \mathbf{I}_2, \quad \mathbf{Ny} = \Sigma_{\Delta} \in \mathbb{R}^{2 \times 2} \quad (62)$$

Continuous time propagation

$$\mathbf{A}(t) = \begin{bmatrix} 0 & 0 & -\hat{u} \sin(\hat{\theta}) & \frac{1}{2} \cos(\hat{\theta}) \tilde{\varphi}_l & \frac{1}{2} \cos(\hat{\theta}) \tilde{\varphi}_r \\ 0 & 0 & \hat{u} \cos(\hat{\theta}) & \frac{1}{2} \sin(\hat{\theta}) \tilde{\varphi}_l & \frac{1}{2} \sin(\hat{\theta}) \tilde{\varphi}_r \\ 0 & 0 & 0 & \frac{1}{L} \tilde{\varphi}_l & -\frac{1}{L} \tilde{\varphi}_r \\ 0 & 0 & 0 & -\lambda_{\delta R_l} & 0 \\ 0 & 0 & 0 & 0 & -\lambda_{\delta R_r} \end{bmatrix} \quad (63)$$

$$\hat{u} = \frac{1}{2}((R_l + \delta \hat{R}_l) \tilde{\varphi}_l + (R_r + \delta \hat{R}_r) \tilde{\varphi}_r)$$

$$\mathbf{B}(t) = \begin{bmatrix} -\frac{1}{2} \cos(\hat{\theta})(R_0 + \delta \hat{R}_l) & -\frac{1}{2} \cos(\hat{\theta})(R_0 + \delta \hat{R}_r) & 1 & 0 & 0 & 0 & 0 \\ -\frac{1}{2} \sin(\hat{\theta})(R_0 + \delta \hat{R}_l) & -\frac{1}{2} \sin(\hat{\theta})(R_0 + \delta \hat{R}_r) & 0 & 1 & 0 & 0 & 0 \\ -\frac{1}{L}(R_0 + \delta \hat{R}_l) & -\frac{1}{L}(R_0 + \delta \hat{R}_r) & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (64)$$

$$\hat{u} = \frac{1}{2}((R_l + \delta \hat{R}_l) \tilde{\varphi}_l + (R_r + \delta \hat{R}_r) \tilde{\varphi}_r)$$

$$\mathbf{Q} = \text{diag}([\sigma_{\dot{\varphi}}^2, \sigma_{\dot{\varphi}}^2, \sigma_n^2, \sigma_e^2, \sigma_{\theta}^2, \sigma_{\delta R}^2, \sigma_{\delta R_l}^2]), \quad \text{such as } \mathbb{E}[\mathbf{w}_k \mathbf{w}_l^T] = \mathbf{Q} \delta_{k-l} \quad (65)$$

A.2 Decentralised approach with ranging measurements

A.2.1 Vehicles WV1,2,3 with GPS

Magnetic field measurements

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & -\sin(\hat{\theta})m_n^t + \cos(\hat{\theta})m_e^t & 0 & 0 & 0 & 0 \\ 0 & 0 & -\cos(\hat{\theta})m_n^t - \sin(\hat{\theta})m_e^t & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{D} = \mathbf{I}_2, \quad \mathbf{Ny} = \sigma_m^2 \cdot \mathbf{I}_2 \quad (66)$$

GPS measurements

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{D} = \mathbf{I}_2, \quad \mathbf{Ny} = \sigma_G^2 \cdot \mathbf{I}_2 \quad (67)$$

Continuous time propagation

$$\mathbf{A}(t) = \begin{bmatrix} 0 & 0 & -\hat{u} \sin(\hat{\theta}) & \frac{1}{2} \cos(\hat{\theta}) \tilde{\varphi}_l & \frac{1}{2} \cos(\hat{\theta}) \tilde{\varphi}_r & 0 & 0 \\ 0 & 0 & \hat{u} \cos(\hat{\theta}) & \frac{1}{2} \sin(\hat{\theta}) \tilde{\varphi}_l & \frac{1}{2} \sin(\hat{\theta}) \tilde{\varphi}_r & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{L} \tilde{\varphi}_l & -\frac{1}{L} \tilde{\varphi}_r & 0 & 0 \\ 0 & 0 & 0 & -\lambda_{\delta R_l} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\lambda_{\delta R_r} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\lambda_{b_{Gn}} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\lambda_{b_{Ge}} \end{bmatrix} \quad (68)$$

$$\hat{u} = \frac{1}{2}((R_l + \delta \hat{R}_l) \tilde{\varphi}_l + (R_r + \delta \hat{R}_r) \tilde{\varphi}_r)$$

$$\mathbf{B}(t) = \begin{bmatrix} -\frac{1}{2} \cos(\hat{\theta})(R_0 + \delta \hat{R}_l) & -\frac{1}{2} \cos(\hat{\theta})(R_0 + \delta \hat{R}_r) & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{2} \sin(\hat{\theta})(R_0 + \delta \hat{R}_l) & -\frac{1}{2} \sin(\hat{\theta})(R_0 + \delta \hat{R}_r) & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{L}(R_0 + \delta \hat{R}_l) & -\frac{1}{L}(R_0 + \delta \hat{R}_r) & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (69)$$

$$\hat{u} = \frac{1}{2}((R_l + \delta \hat{R}_l) \tilde{\varphi}_l + (R_r + \delta \hat{R}_r) \tilde{\varphi}_r)$$

$$\mathbf{Q} = \text{diag}([\sigma_{\tilde{\varphi}}^2, \sigma_{\tilde{\varphi}}^2, \sigma_n^2, \sigma_e^2, \sigma_{\hat{\theta}}^2, \sigma_{\delta R}^2, \sigma_{\delta R}^2, \sigma_{b_G}^2, \sigma_{b_G}^2]), \quad \text{such as } \mathbb{E}[\mathbf{w}_k \mathbf{w}_l^T] = \mathbf{Q} \delta_{k-l} \quad (70)$$

A.2.2 Vehicle WV4 without GPS

Magnetic field measurements

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & -\sin(\hat{\theta})m_n^t + \cos(\hat{\theta})m_e^t & 0 & 0 & \mathbf{0}^{1 \times 6} \\ 0 & 0 & -\cos(\hat{\theta})m_n^t - \sin(\hat{\theta})m_e^t & 0 & 0 & \mathbf{0}^{1 \times 6} \end{bmatrix}, \quad \mathbf{D} = \mathbf{I}_2, \quad \mathbf{Ny} = \sigma_m^2 \cdot \mathbf{I}_2 \quad (71)$$

Ranging measurements

$$\mathbf{C}(t) \in \mathbb{R}^{3 \times 11} \Rightarrow \begin{cases} \mathbf{C}_{1:3,1:2} = \begin{bmatrix} \frac{\hat{n}-\hat{n}_1}{\sqrt{(\hat{n}-\hat{n}_1)^2+(\hat{e}-\hat{e}_1)^2}} & \frac{\hat{e}-\hat{e}_1}{\sqrt{(\hat{n}-\hat{n}_1)^2+(\hat{e}-\hat{e}_1)^2}} \\ \frac{\hat{n}-\hat{n}_2}{\sqrt{(\hat{n}-\hat{n}_2)^2+(\hat{e}-\hat{e}_2)^2}} & \frac{\hat{e}-\hat{e}_2}{\sqrt{(\hat{n}-\hat{n}_2)^2+(\hat{e}-\hat{e}_2)^2}} \\ \frac{\hat{n}-\hat{n}_3}{\sqrt{(\hat{n}-\hat{n}_3)^2+(\hat{e}-\hat{e}_3)^2}} & \frac{\hat{e}-\hat{e}_3}{\sqrt{(\hat{n}-\hat{n}_3)^2+(\hat{e}-\hat{e}_3)^2}} \end{bmatrix} \\ \mathbf{C}_{1,6} = c \\ \mathbf{C}_{2,8} = c \\ \mathbf{C}_{3,10} = c \\ \mathbf{C}_{i,j} = 0 \quad (\text{everywhere else}) \end{cases}, \quad \mathbf{D} = \mathbf{I}_6, \quad \mathbf{N}_y = \Sigma_\rho \quad (72)$$

Continuous time propagation

$$\mathbf{A}(t) \in \mathbb{R}^{11 \times 11} \Rightarrow \begin{cases} \mathbf{A}_{1:5,1:5} = \begin{bmatrix} 0 & 0 & -\hat{u} \sin(\hat{\theta}) & \frac{1}{2} \cos(\hat{\theta}) \tilde{\varphi}_l & \frac{1}{2} \cos(\hat{\theta}) \tilde{\varphi}_r \\ 0 & 0 & \hat{u} \cos(\hat{\theta}) & \frac{1}{2} \sin(\hat{\theta}) \tilde{\varphi}_l & \frac{1}{2} \sin(\hat{\theta}) \tilde{\varphi}_r \\ 0 & 0 & 0 & \frac{1}{L} \tilde{\varphi}_l & -\frac{1}{L} \tilde{\varphi}_l \\ 0 & 0 & 0 & -\lambda_{\delta R_l} & 0 \\ 0 & 0 & 0 & 0 & -\lambda_{\delta R_r} \end{bmatrix} \\ \mathbf{A}_{6,7} = 1 \\ \mathbf{A}_{8,9} = 1 \\ \mathbf{A}_{10,11} = 1 \\ \mathbf{A}_{i,j} = 0 \quad (\text{everywhere else}) \end{cases} \quad (73)$$

$$\hat{u} = \frac{1}{2}((R_l + \delta \hat{R}_l) \tilde{\varphi}_l + (R_r + \delta \hat{R}_r) \tilde{\varphi}_r)$$

$$\mathbf{B}(t) \in \mathbb{R}^{11 \times 10} \Rightarrow \begin{cases} \mathbf{B}_{1:3,1:2} = \begin{bmatrix} -\frac{1}{2} \cos(\hat{\theta})(R_0 + \delta \hat{R}_l) & -\frac{1}{2} \cos(\hat{\theta})(R_0 + \delta \hat{R}_r) \\ -\frac{1}{2} \sin(\hat{\theta})(R_0 + \delta \hat{R}_l) & -\frac{1}{2} \sin(\hat{\theta})(R_0 + \delta \hat{R}_r) \\ -\frac{1}{L}(R_0 + \delta \hat{R}_l) & -\frac{1}{L}(R_0 + \delta \hat{R}_r) \end{bmatrix} \\ \mathbf{B}_{1:5,3:7} = \mathbf{I}_5 \\ \mathbf{B}_{7,8} = 1 \\ \mathbf{B}_{9,9} = 1 \\ \mathbf{B}_{11,10} = 1 \\ \mathbf{B}_{i,j} = 0 \quad (\text{everywhere else}) \end{cases} \quad (74)$$

$$\hat{u} = \frac{1}{2}((R_l + \delta \hat{R}_l) \tilde{\varphi}_l + (R_r + \delta \hat{R}_r) \tilde{\varphi}_r)$$

$$\mathbf{Q} = \text{diag}([\sigma_{\tilde{\varphi}}^2, \sigma_{\tilde{\varphi}}^2, \sigma_n^2, \sigma_e^2, \sigma_\theta^2, \sigma_{\delta R}^2, \sigma_{\delta R}^2, \sigma_f^2, \sigma_f^2, \sigma_f^2]), \quad \text{such as} \quad \mathbb{E}[\mathbf{w}_k \mathbf{w}_l^T] = \mathbf{Q} \delta_{k-l} \quad (75)$$

A.3 Centralised approach

Magnetic field measurements

$$\mathbf{C}(t) \in \mathbb{R}^{8 \times 38} \Rightarrow \begin{cases} \mathbf{C}_{1:2,3} = \begin{bmatrix} -\sin(\hat{\theta}_1)m_{n,1}^t + \cos(\hat{\theta}_1)m_{e,1}^t \\ -\cos(\hat{\theta}_1)m_{n,1}^t - \sin(\hat{\theta}_1)m_{e,1}^t \end{bmatrix} \\ \mathbf{C}_{3:4,10} = \begin{bmatrix} -\sin(\hat{\theta}_2)m_{n,2}^t + \cos(\hat{\theta}_2)m_{e,2}^t \\ -\cos(\hat{\theta}_2)m_{n,2}^t - \sin(\hat{\theta}_2)m_{e,2}^t \end{bmatrix} \\ \mathbf{C}_{5:6,17} = \begin{bmatrix} -\sin(\hat{\theta}_3)m_{n,3}^t + \cos(\hat{\theta}_3)m_{e,3}^t \\ -\cos(\hat{\theta}_3)m_{n,3}^t - \sin(\hat{\theta}_3)m_{e,3}^t \end{bmatrix} \\ \mathbf{C}_{7:8,24} = \begin{bmatrix} -\sin(\hat{\theta}_4)m_{n,4}^t + \cos(\hat{\theta}_4)m_{e,4}^t \\ -\cos(\hat{\theta}_4)m_{n,4}^t - \sin(\hat{\theta}_4)m_{e,4}^t \end{bmatrix} \\ \mathbf{C}_{i,j} = 0 \quad (\text{everywhere else}) \end{cases}, \quad \mathbf{D} = \mathbf{I}_8, \quad \mathbf{N}\mathbf{y} = \sigma_m^2 \cdot \mathbf{I}_8 \quad (76)$$

GPS measurements

$$\mathbf{C}(t) \in \mathbb{R}^{6 \times 38} \Rightarrow \begin{cases} \mathbf{C}_{1:2,1:7} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \\ \mathbf{C}_{3:4,8:14} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \\ \mathbf{C}_{5:6,15:21} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \\ \mathbf{C}_{i,j} = 0 \quad (\text{everywhere else}) \end{cases}, \quad \mathbf{D} = \mathbf{I}_6, \quad \mathbf{N}\mathbf{y} = \sigma_G^2 \cdot \mathbf{I}_6 \quad (77)$$

Ranging measurements

$$\mathbf{C} \in \mathbb{R}^{20 \times 38} \Rightarrow \begin{cases} \mathbf{C}_{\mathbf{k}(i)+(j-i),7(j-1)+1} = \frac{\hat{n}_j - \hat{n}_i}{\hat{\rho}_{i,j}}, & (i, j = 1, \dots, 4 \mid i < j) \\ \mathbf{C}_{\mathbf{k}(i)+(j-i),7(j-1)+2} = \frac{\hat{e}_j - \hat{e}_i}{\hat{\rho}_{i,j}}, & (i, j = 1, \dots, 4 \mid i < j) \\ \mathbf{C}_{\mathbf{k}(i)+(j-i),7(i-1)+1} = \frac{\hat{n}_i - \hat{n}_j}{\hat{\rho}_{i,j}}, & (i, j = 1, \dots, 4 \mid i < j) \\ \mathbf{C}_{\mathbf{k}(i)+(j-i),7(i-1)+2} = \frac{\hat{e}_i - \hat{e}_j}{\hat{\rho}_{i,j}}, & (i, j = 1, \dots, 4 \mid i < j) \\ \mathbf{C}_{\mathbf{k}(i)+(j-i),26+2(\mathbf{k}(i)-6)} = c, & (i, j = 1, \dots, 4 \mid i < j) \\ \mathbf{C}_{i,j} = 0 & (\text{everywhere else}) \end{cases}, \quad \mathbf{D} = \mathbf{I}_6, \quad \mathbf{N}\mathbf{y} = \Sigma_\rho \in \mathbb{R}^{6 \times 6} \quad (78)$$

with $k = [0 \ 3 \ 5]$ and $\hat{\rho}_{i,j} = \sqrt{(\hat{n}_j - \hat{n}_i)^2 + (\hat{e}_j - \hat{e}_i)^2}$

Continuous time propagation

$$\mathbf{A}(t) \in \mathbb{R}^{38 \times 38} \implies \begin{cases} A_{7(i-1)+1,7(i-1)+3} = -\hat{u}_i \sin(\theta_i), & (i = 1, \dots, 4) \\ A_{7(i-1)+1,7(i-1)+4} = \frac{1}{2} \cos(\theta_i) \tilde{\varphi}_{l,i}, & (i = 1, \dots, 4) \\ A_{7(i-1)+1,7(i-1)+5} = \frac{1}{2} \cos(\theta_i) \tilde{\varphi}_{r,i}, & (i = 1, \dots, 4) \\ A_{7(i-1)+2,7(i-1)+3} = \hat{u}_i \cos(\theta_i), & (i = 1, \dots, 4) \\ A_{7(i-1)+2,7(i-1)+4} = \frac{1}{2} \sin(\theta_i) \tilde{\varphi}_{l,i}, & (i = 1, \dots, 4) \\ A_{7(i-1)+2,7(i-1)+5} = \frac{1}{2} \sin(\theta_i) \tilde{\varphi}_{r,i}, & (i = 1, \dots, 4) \\ A_{7(i-1)+3,7(i-1)+4} = \frac{1}{L} \tilde{\varphi}_{l,i}, & (i = 1, \dots, 4) \\ A_{7(i-1)+3,7(i-1)+5} = -\frac{1}{L} \tilde{\varphi}_{r,i}, & (i = 1, \dots, 4) \\ A_{7(i-1)+4,7(i-1)+4} = -\lambda_{\delta R_l}, & (i = 1, \dots, 4) \\ A_{7(i-1)+5,7(i-1)+5} = -\lambda_{\delta R_r}, & (i = 1, \dots, 4) \\ A_{7(i-1)+6,7(i-1)+6} = -\lambda_G, & (i = 1, 2, 3) \\ A_{7(i-1)+7,7(i-1)+7} = -\lambda_G, & (i = 1, 2, 3) \\ A_{k(i)+2(j-i),k(i)+2(j-i)} = 1, & (i, j = 1, \dots, 4 \mid i < j) \end{cases} \quad (79)$$

with $k = [26 \ 32 \ 36]$ and $\hat{u}_i = \frac{1}{2}((R_l + \delta \hat{R}_{l,i}) \tilde{\varphi}_{l,i} + (R_r + \delta \hat{R}_{r,i}) \tilde{\varphi}_{r,i})$

$$\mathbf{B}(t) \in \mathbb{R}^{38 \times 40} \implies \begin{cases} B_{7(i-1)+1,9(i-1)+1} = -\frac{1}{2} \cos(\hat{\theta}_i)(R_0 + \delta \hat{R}_{l,i}), & (i = 1, \dots, 4) \\ B_{7(i-1)+1,9(i-1)+2} = -\frac{1}{2} \cos(\hat{\theta}_i)(R_0 + \delta \hat{R}_{r,i}), & (i = 1, \dots, 4) \\ B_{7(i-1)+2,9(i-1)+1} = -\frac{1}{2} \sin(\hat{\theta}_i)(R_0 + \delta \hat{R}_{l,i}), & (i = 1, \dots, 4) \\ B_{7(i-1)+2,9(i-1)+2} = -\frac{1}{2} \sin(\hat{\theta}_i)(R_0 + \delta \hat{R}_{r,i}), & (i = 1, \dots, 4) \\ B_{7(i-1)+3,9(i-1)+1} = -\frac{1}{L}(R_0 + \delta \hat{R}_{l,i}), & (i = 1, \dots, 4) \\ B_{7(i-1)+3,9(i-1)+3} = -\frac{1}{L}(R_0 + \delta \hat{R}_{r,i}), & (i = 1, \dots, 4) \\ B_{7(i-1)+(1:5),9(i-1)+(3:7)} = \mathbf{I}_5, & (i = 1, \dots, 4) \\ B_{7(i-1)+(6:7),9(i-1)+(8:9)} = \mathbf{I}_2, & (i = 1, 2, 3) \\ B_{k(i)+2(j-i),k2(i)+j-i} = 1, & (i, j = 1, \dots, 4 \mid i < j) \end{cases} \quad (80)$$

with $k = [26 \ 32 \ 36]$ and $k2 = [34 \ 37 \ 39]$

$$\mathbf{Q} = \text{diag}([\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \mathbf{q}_4, \sigma_f^2, \sigma_f^2, \sigma_f^2, \sigma_f^2, \sigma_f^2]), \quad \text{such as } \mathbb{E}[\mathbf{w}_k \mathbf{w}_l^T] = \mathbf{Q} \delta_{k-l} \quad (81)$$

with

$$\begin{aligned} \mathbf{q}_i &= [\sigma_{\varphi}^2, \sigma_{\varphi}^2, \sigma_n^2, \sigma_e^2, \sigma_{\theta}^2, \sigma_{\delta R}^2, \sigma_{\delta R}^2, \sigma_G^2, \sigma_G^2], & (i = 1, 2, 3) \\ \mathbf{q}_4 &= [\sigma_{\varphi}^2, \sigma_{\varphi}^2, \sigma_n^2, \sigma_e^2, \sigma_{\theta}^2, \sigma_{\delta R}^2, \sigma_{\delta R}^2]. \end{aligned}$$

Magnetic field at Trudeau airport (Montreal, Canada) : <https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml#igrfwmm>

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